

$$X \sim f_X(x)$$

$$x_1 \dots x_n$$

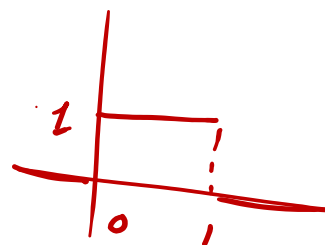
$$\underline{Z} \sim \frac{g(z)}{Z} \quad \mathcal{N}(0, I)$$

$$Y = h(Z) \sim f_X(x)$$

$$f_Y(y) \propto g_Z(h^{-1}(y))$$

$$\text{ex} \quad Z \sim U(-1, 1)$$

$$Y = Z^2$$



$$P(Y \leq y) = P(Z^2 \leq y) = P(|Z| \leq \sqrt{y})$$

$$f_Y(y) = \begin{cases} \frac{1}{2\sqrt{y}} & y \geq 0 \\ 0 & y < 0 \end{cases}$$

$$= P(-\sqrt{y} \leq Z \leq \sqrt{y}) = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2} dz$$

$$= \sqrt{y} \quad y \geq 0$$

Process

$$X_0 \sim N(0, I)$$

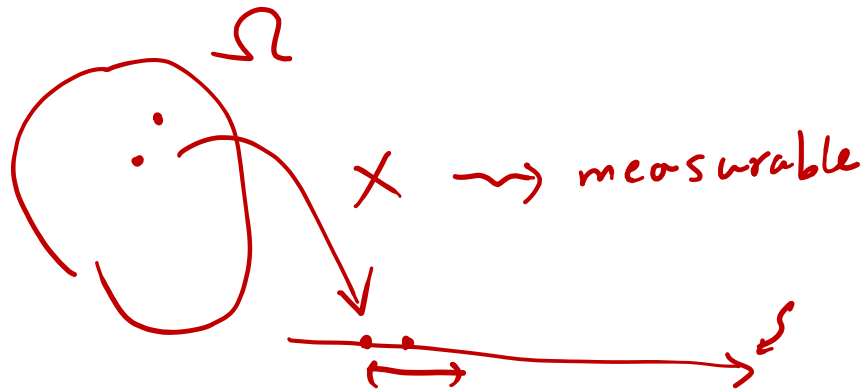
Stochastic Process

$$X_t$$

$$0 \leq t \leq 1$$

$$\begin{cases} X_0 \sim N(0, I) \\ X_1 \sim f_x \end{cases}$$

Random Variable



Prob. Space

$$(\Omega, \underline{F}, P)$$

Event

$$F \subseteq 2^\Omega$$

$$F \text{ } \sigma\text{-Algebra}$$

- ① $\Omega \in F$
- ② $A \in F \Rightarrow A^c \in F$
- ③ $\forall i A_i \in F, \bigcup_{i=1}^{\infty} A_i \in F$

$$\Omega = \{1, \dots, 6\}$$

$$F = \{\{\}, \{1, \dots, 6\}, \{2\}\} \quad \text{is it } \sigma\text{-algebra?}$$

$$F = \{\{\}, \{1, \dots, 6\}, \{2\}, \{1, 3, 4, 5, 6\}\}$$

$$P: F \rightarrow [0, 1]$$

$$\textcircled{1} \quad P(\Omega) = 1$$

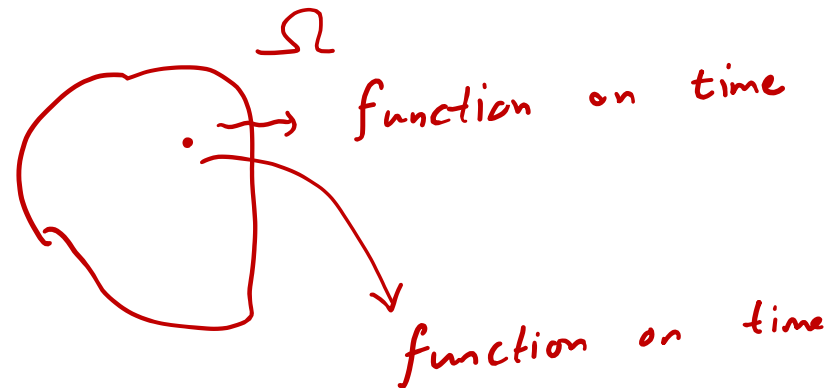
$$\textcircled{2} \quad P(A \cup B) = P(A) + P(B)$$

$$A \cap B = \emptyset$$

$$\textcircled{3} \quad P(A^c) = 1 - P(A)$$

$$\text{Density} \quad P(I) = \int_I \widehat{f}_X(x) dx$$

Random Experiment :



How to characterize the prob. measure in stochastic process?

$$\{ f_{x_t}(x; t) \quad 0 \leq t \leq 1 \}$$

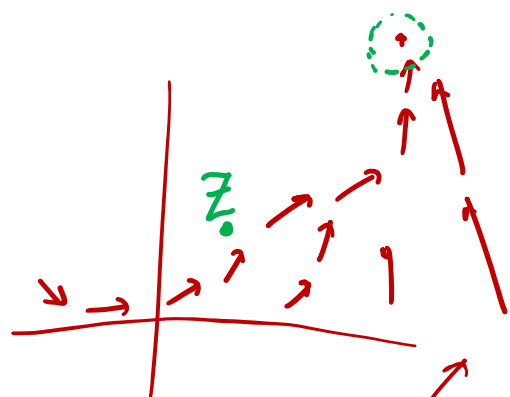
$$\{ F_{x_t}(x; t) \quad 0 \leq t \leq 1 \}$$

$$\forall k \quad \{ f_{x_{t_1}, x_{t_2}, \dots, x_{t_k}}(x_1, \dots, x_k) \}$$

* Sampling of Stochastic Processes

ODE / SDE

u Continuously Differ.



$$\Phi(t, \vec{x}_0) = \vec{x}_t$$

ODE

$$\frac{dX(t)}{dt} = u(X(t)) \quad 0 \leq t \leq 1$$

deterministic

$$\underline{X(\cdot)} = \underline{Z}$$

$$X(t) \in \mathbb{R}^d$$

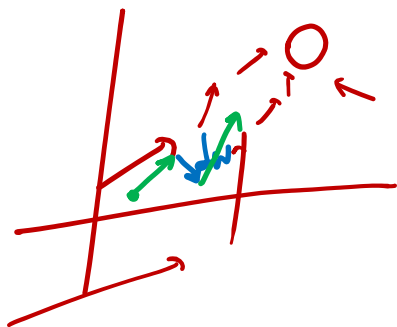
$$\underline{x_t} \rightarrow \boxed{\theta} \rightarrow \underline{u_{\theta}(x_t)}$$

$$\frac{X(t+h) - X(t)}{h} = u(X(t))$$

$$X(t+h) = X(t) + h u(\underline{X(t)}) \quad \text{Euler}$$

$$X(1) \sim f_X \quad x_1, \dots, x_n$$

$$\min_{\theta} \mathbb{E} \{ u_{\text{target}} - u_{\theta} \}^2$$



SDE

$$X(t+h) = X(t) + \underbrace{h \downarrow u(X_t)}_{\text{step size}} + \underbrace{\sigma_t dW}_{\mathcal{N}(0, h)} \quad \epsilon_t \sim \mathcal{N}(0, 1)$$

Brownian Motion

$$W_t$$

$$W_0 = 0$$

$$W_s - W_t$$

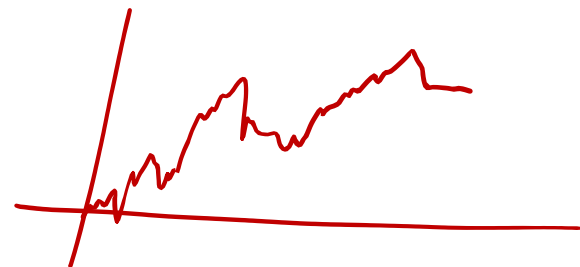
$$\sim \mathcal{N}(0, (s-t))$$

$$t_1 < t_2 < \dots < t_n$$

$$W_{t_2} - W_{t_1},$$

$$W_{t_3} - W_{t_2}, \dots, W_{t_n} - W_{t_{n-1}}$$

independent



Continuously
Diff.

